

Answer Key — Stacks, Queues & the Public/Private Divide

Instructor reference for the review worksheet

array[0] Fill in the Blank

1. class → **private** · struct → **public**
2. **new**
3. **top**
4. **push** · **pop**
5. **rear** · **front**
6. **encapsulation**
7. **abstraction**
8. **private**
9. **public** (label before `void push(...)`)

array[1] Find the Error

Case A Does not compile. `Stack(int size)` sits above the `public:` label, so it's private by default. `Stack S(10);` in `main()` calls a private constructor from outside the class — `error: 'Stack::Stack(int)' is private within this context`.
DOES NOT COMPILE

Case B `top` is declared but never assigned in the constructor (only `array` is set). It starts as an indeterminate/garbage value, so the first `push()` writes to `array[ ++top ]` at an unpredictable — possibly out-of-bounds — index.
COMPILES, UNDEFINED BEHAVIOR

Case C `front` is a private member of `Queue`. `main()` is outside the class, so `Q.front = nullptr;` is illegal no matter how harmless the assignment looks — access is checked at compile time, not by intent.
DOES NOT COMPILE

Case D Missing semicolon after the closing brace of the `Node` class definition. Without it, the compiler reads the class body and `int main()` as one run-on declaration.
DOES NOT COMPILE

array[2] Short Answer — model answers

1. *Encapsulation vs. abstraction*: Encapsulation bundles an object's data with the functions that operate on it and restricts outside access — e.g., making `array` and `top` private in `Stack`. Abstraction exposes only the operations a caller needs (`push()`, `pop()`, `print()`) while hiding whether the implementation is an array or a linked list.
2. *front/rear vs. top*: A stack only ever adds and removes at one end, so a single `top` index locates that end. A queue adds at one end and removes at the other, so it needs two references — `front` and `rear` — to reach both ends in constant time.
3. *Broken invariant example*: Any direct write with no bounds check, e.g. `S.top = 999;` or `S.array[50] = 42;` — either desyncs `top` from the array's real contents.
4. *Fixing Case A*: Move the constructor under `public:`, or add a separate public constructor/factory function that `main()` can call.
5. *LIFO / FIFO*: LIFO ("last in, first out") describes the Stack — the most recently pushed value is the first one popped. FIFO ("first in, first out") describes the Queue — the earliest value pushed is the first to leave.